# Grace Wilding

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## Education

**Case Western Reserve University School of Medicine** — Cleveland, OH
*M.S. in Biostatistics (concurrent with B.S.)* — *Aug 2026 – May 2028*

- **Current Coursework**: Statistical Methods I, Data Management & Statistical Programming (SAS); by May 2027: Statistical Methods II, Survival Analysis, Clinical Trials, Research Seminar

**Case Western Reserve University** — Cleveland, OH
*B.S. in Applied Mathematics & Statistics | GPA: 4.0/4.0* — *Aug 2024 – May 2028*

- **Honors:** Case Alumni Association Junior Senior Scholarship (2026, selective merit award for STEM majors); Dean's High Honor List
- **Coursework**: Probability, Statistical Inference, Data Analysis & Linear Models, Math in Machine Learning, Real Analysis I, Scientific Computing, Linear Algebra, Differential Equations, Calculus I - III, Mathematical Modeling, Data Structures

## Research Experience

**Biostatistics Research Assistant** — Aug 2026 – Present
*University of Florida, College of Public Health & Health Professions* — *Remote*

- Fit multi-state Markov models to estimate transition probabilities between Alzheimer's disease stages and derive cumulative conversion and survival estimates (co-author, manuscript under review at J. Alzheimer's Disease)
- Designed a Monte Carlo simulation study in R (5,000 replications) on data mimicking the Phase 3 donanemab trials to compare selection frequencies of competing non-nested longitudinal models
- Conducted the literature review and authored the Introduction and Discussion of a manuscript on non-nested model selection for Alzheimer's disease clinical trials (submitted)

**Biostatistics Research Intern – Mentor: Dr. Yajun Mei** — Jun 2026 – Jul 2026
*New York University School of Global Public Health – NIH/NIA-funded summer program* — *New York, NY*

- Built a Future Elderly Model (logistic transition models within a Markov microsimulation framework) using Health and Retirement Study (HRS) panel data to project life expectancy and chronic disease prevalence
- Cleaned and processed HRS data, engineered predictors, fit logistic models, and ran counterfactual smoking-policy simulations in R
- Presented "Smoking, Life Expectancy and Chronic Disease in the United States" at the PQAR Annual Symposium; completed program coursework in biostatistics, epidemiology, and statistical computing

**Research Assistant, Sahoo Lab (Biomedical & Health Informatics)** — Apr 2025 – May 2026
*Case Western Reserve University School of Medicine* — *Cleveland, OH*

- Developed backend functionality and data organization in Python/Django for a web platform aggregating Dravet Syndrome literature for clinicians and the public
- Helped integrate an AI chatbot built on a literature-derived knowledge graph; worked with clinical researchers to translate findings into an accessible tool

## Publications

Qin, S., **Wilding, G.**, Bernick, C., & Shan, G. (2026). Cumulative transition probabilities from multi-state Markov models to analyze conversion from healthy normal to Alzheimer's disease dementia. *Journal of Alzheimer's Disease* (revised, under review).

Ding, A., **Wilding, G.**, Tang, Z., & Shan, G. (2026). Non-nested model selection in Alzheimer's disease research. Manuscript submitted for publication.

## Projects

**High Dimensional Probability & Asymptotic Geometry of Sequence Spaces** Sept 2026 - Dec 2026

- Studying concentration of measure and its application to convex geometry, including asymptotic behavior in $l_p$ and Lorentz spaces; presenting to the CWRU Mathematics Department in December 2026

**Stochastic Model of Optimal Foraging – MATH 358** April 2026

- Modeled predator encounters and travel times as Poisson/exponential processes; derived analytical survival thresholds and ran optimization, sensitivity analysis, and long-run simulations in MATLAB

**SIR Epidemic Model with Vaccination & Demographics – MATH 358** February 2026

- Computed equilibria and analyzed the role of $R_0$ in disease persistence; numerical simulation and vector-field visualization in MATLAB

**Graph-Theoretic Modeling of Disease Transmission** December 2025

- Simulated network-based disease spread and evaluated how modeling choices affect outcomes; presented to the CWRU Mathematics Department

## Technical Skills

**Languages**: R, SAS, Python (Django), Java
**Statistical Methods**: Multi-state Markov models, logistic regression, linear models, Monte Carlo simulation studies, microsimulation, model selection, transition-probability and survival estimation, stochastic processes
**Data & Tools**: Longitudinal/panel survey data (HRS), data cleaning and management, exploratory analysis, visualization, RStudio, Excel

## Teaching, Leadership & Service

**Biostatistics M.S. Student Representative** — CWRU School of Medicine Sept 2026 - Present

- Serve as a liaison between graduate students and program leadership; plan and organize student activities

**Grader, Modern Physics** — CWRU Department of Physics Sept 2026 - Present

- Grade weekly assignments and exams in coordination with the instructor and grading team; proctor exams

**Vice President of Speaker Relations** — TEDxCWRU Aug 2025 - Present

- Recruit conference speakers aligned with the event theme and lead speaker preparation sessions

**Vice President of Administration & Outreach** — National Residence Hall Honorary Jan 2025 - Present

- Build and track the annual budget; manage the membership roster, social media, and recruitment materials

**Executive Assistant** — Kappa Alpha Theta, Eta Pi Feb 2025 - Present

- Track attendance and membership requirements; coordinate communication between members and the executive board

**CCEL Scholar** — CWRU Center for Civic Engagement & Learning Aug 2024 - Present

- Complete 45+ volunteer hours per year with organizarions including the Cleveland Food Bank and Cleveland Book Bank